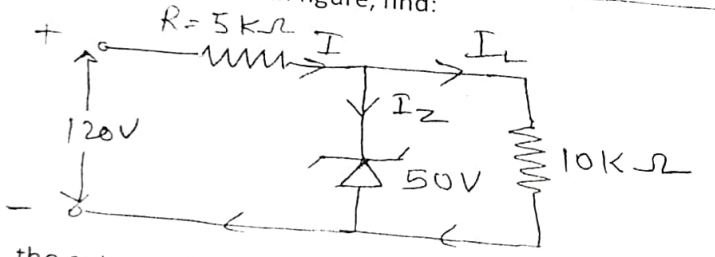
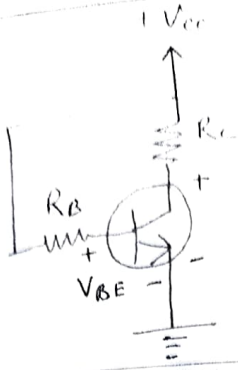
(Please write your Enrollment Number) Enrollment No. 02701182021

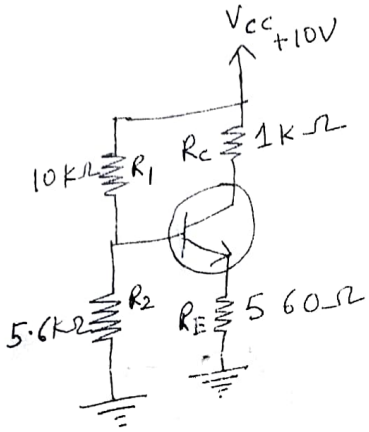
MID-TERM EXAMINATION
(Course Name: B. Tech) (Semester: I)
(October, 2025) OFF LINE mode

Subject Code: BEC 103	Subject: Electronics Workshop
Time : 1 ½Hours	Maximum Marks :30

Note: Q1 is compulsory. Attempt any two parts of Q2 and Q3

Q1		(2.5*4=10)	CO Mapping
	a) Draw VI characteristics of a PN junction diode and explain it's working in forward biased mode.	2.5	1
	b) A heavily doped n type semiconductor has the following data: hole electron mobility ratio: 0.4, doping concentration: 4.2×10^8 atoms/m^3. Calculate the conductance of the n type semiconductor.	2.5	1
	c) What are absolute and secondary measuring instruments? Give example.	2.5	2
	d) What are the characteristics of measuring instruments?	2.5	2
	UNIT I		
Q2	**Attempt any two parts**	**(5*2=10)**	**CO Mapping**
	a) With a neat diagram, explain the working of a full wave bridge rectifier.	5	1
	b) What are the different modes of operation of a transistor? Explain the working of a p-n-p transistor under active mode.	5	1
	c) For the circuit shown in figure, find:  (i) the output voltage, (ii) the voltage drop across series resistance, (iii) current through Zener diode.	5	1
	UNIT II		
Q3	**Attempt any two parts**	**(5*2=10)**	**CO Mapping**
	a) Calculate the operating point for the transistor shown in figure. Assume V_{CC} = 8 Volt, R_B = 360 KΩ, β= 100, R_C = 2 KΩ. Draw DC load line and find the operating point.	5	2



b) Determine Q point for the voltage divider biased circuit shown in figure. Given $\beta = 100$ and $I_C \approx I_E$.		5	2
c) Define and explain, with examples, the different types of possible errors in measurement.		5	2

2/2